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The Measure of All Things

Two great theories shook the world and shattered the earth in the first quarter of the twentieth century, scattering the remains of the physics that had come before and forever altering our understanding of reality. One of these theories, relativity, was developed in true science-fiction fashion, by a lone genius working in splendid isolation, who had left the academy only to return triumphant with profound truth in his hand—this was, of course, Albert Einstein.

The other theory, quantum physics, had a more difficult birth. It was a collaborative effort involving dozens of physicists working over the course of nearly thirty years. Einstein was among them, but he was not their leader; the closest thing this disorganized and unruly band of revolutionaries had was Niels Bohr, the great Danish physicist. Bohr's Institute for Theoretical Physics in Copenhagen was the mecca of quantum physics in its infancy, with nearly every big name in the field for fifty years studying there at one point or another. The physicists who worked there made profound discoveries across nearly every field of science: they developed the first genuine theory of quantum physics, found the underlying logic of the periodic table of the elements, and used the power of radioactivity to reveal the basic workings of living cells. And it was Bohr, along with a group of his most talented students and colleagues—Werner Heisenberg, Wolfgang Pauli, Max Born, Pascual Jordan, and others—who developed and championed the "Copenhagen interpretation," which rapidly became the standard interpretation of the mathematics of quantum physics. What does